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LOAD BALANCE OPTIMIZATION IN PEER CLASSIFIER ROBIN METHOD AS HYBRID FROM PEER CONNECTION CLASSIFIER AND ROUND ROBIN METHODS

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Abstract

Internet usage is currently getting higher at this time. It can increase bandwidth requirements and a build-up of the gateway so that it can make in slow internet access. Load balance is a technique for distributing traffic loads on two or more connection lines in a balanced manner. Then, the traffic can run optimally, maximize throughput, reduce delay time and minimize overloading one of the connection lines. Some methods of applying the principle of load balance are ECMP, Static Route, Round Robin, Peer Connection Classifier (PCC). This paper offers a new way of load balance, namely Peer Classifier Robin (PCR), so Peer Classifier Robin (PCR) is a hybrid of the Peer Connection Classifier and Round Robin methods. The results showed that PCR is a modification of Peer Connection Classifier (PCC), and Round Robin (RR) can be a solution for load balancing because it can optimally minimize weaknesses in Peer Connection Classifier (PCC) and Round Robin (RR). The calculation of the Wireshark shows the PCR method can reduce latency (1.94 ms) and increase throughput (2.32 Mbps) when accessing the internet.

Keywords: Load balance, Peer connection classifier, Peer robin classifier, Round Robin.

1. Introduction

The internet is a global system of computer networks that are interconnected with one another by using the Internet Protocol Suite (Transmission Control Protocol / Internet Protocol) standard to serve users throughout the world. The development of the internet in the current global era is very rapid because users today are very dependent on the internet in solving their daily lives [1].

The problem of companies or users usually found when accessing the internet is a spike in traffic due to unbalanced traffic loads, so the distribution of traffic load on the connection line is not balanced. It can impact on decreasing performance of company performance, such as slow information exchange, unstable application usage, and others [2]. Some of the solutions offered so far are applying the load balance method, so the traffic runs in a balanced and optimal manner.

Load balance is a technique for distributing traffic loads on two or more connection lines in a balanced manner so that it can run optimally, maximize throughput, reduce delay time and minimize overloading one of the connection lines. Regular load balances are applied by users with two internet providers or commonly called ISPs. The condition is useful for optimizing traffic when accessing the internet. Load balance makes a load of traffic balanced on two different gateway paths [3, 4].

Several surveys work of the load balancing methods has been presented [5, 6]. Every survey was focusing on a distinctive point of view. Some points of view are energy requirements [7], distributed systems [8], cloud computing [9, 10], or cloud environments [11-13]. Generally, the selected techniques are compared with each other considering some important qualitative parameters such as fault-tolerant, throughput, energy consumption, heterogeneity, latency or delay time, overhead, packet delivery ratio, packet loss ratio, network lifetime, reliability, and scalability [5]. The load balancing technique has advantages and disadvantages in several parameters, respectively. Some of the methods are the Round Robin method, the Peer Connection Classifier, the Static route, Honey Bee algorithm [14], and Equal-Cost Multi-Path (ECMP) [6, 15, 16], machine learning approach [17] or optimization techniques [18]. The paper efforts on analysing two load balancing techniques, namely the Round Robin method and the Peer Connection Classifier (PCC). The Round Robin algorithm serves packet requests sequentially based on queues [19]. The use of the Round Robin method is always not fixed and randomly based on browsing and downloading sessions [20]. It can affect to guaranteeing the relationship between the server and the client because the gateway used is not always the same when accessing the internet. As a result, when accessing the internet, it is possible that the internet feels slow when we browse several different websites at the same time [21, 22]. Whereas the PCC algorithm works based on packet traffic on the gateway path that is still interrelated so that the packet will be missed. PCC itself also occurs a high delay due to traffic congestion in one of the gateway lines. At the same time, the static route method is a method where the algorithm determines one IP range to be passed by a particular gateway [23, 24].

This study focused on two methods that are used by users, namely PCC and Round Robin. It will be tested and compared by the modified method, namely Peer Classifier Robin (PCR). PCR is a technique born from the modified PCC and Round Robin method. This PCR method is intended to occur in the process optimization in the implementation of the load balance principle. The testing

carried out in the Laboratory with QoS measurement parameters, such as Packet Loss, Delay Time, Throughput, and usage traffic using Wireshark. Quality of Service is used to measure the quality level of the internet or intranet TCP/IP network connections.

Therefore, the paper analyses the PCC, Round Robin, and Peer Classifier Robin (PCR) methods aimed at testing and comparing the appropriate techniques to be applied in the agency network. Also, this article discusses providing a general description of optimizing load balance in internet usage. The application of the Peer Classifier Robin (PCR) method is expected to optimize the load balancing process in the implementation of internet-based communication.

2. Load Balance

Load balance is a technique to move or distribute traffic when accessing the internet on two connection lines, evenly load balance is a technique to run or distribute traffic when accessing the internet on two connection lines evenly. It aims to minimize overloading the connection lines, maximize the occurrence of throughput, and reduce the delay time in traffic usage.

The load balance does not make the one connection on two providers so that it becomes flat. Still, load balance only distinguishes the gateway that traffic traverses when accessing the internet, so it is expected that there will be no build-up of traffic on the wrong path [19, 20].

Figure 1 is a load balancing process using several supporting equipment such as a server, router, internet provider. The router plays a role while implementing a method that has been configured in advance. The primary purpose of all existing load balancing methods is to distribute the load evenly on the connection path. When the load balance method receives a service request from the user, then the application will be forwarded to the main server. Usually, load balance methods can smartly determine which server has a lower load and faster response. It can even stop access to servers that are experiencing problems and only forward them to servers that can provide services.

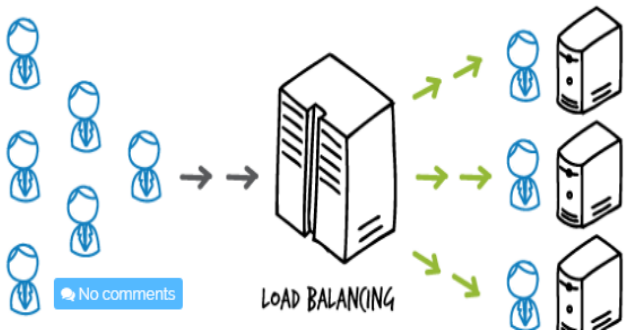


Fig. 1. Load balancing process.

2.1. PCC (Per Connection Classifier)

The PCC method is a load balance method that uses an algorithm where the relationship between server and client will be guaranteed because when accessing

the internet, traffic is passed to the same connection path if the data packet is still interrelated. Figure 2 shows the workings of the PCC method while makes several groups of connection traffic that pass through the router path so that the router will know the gateway path. Then, the path is passed at the beginning of the connection traffic, and the next packages that are still related to the connection will initially be passed on the same gateway too.

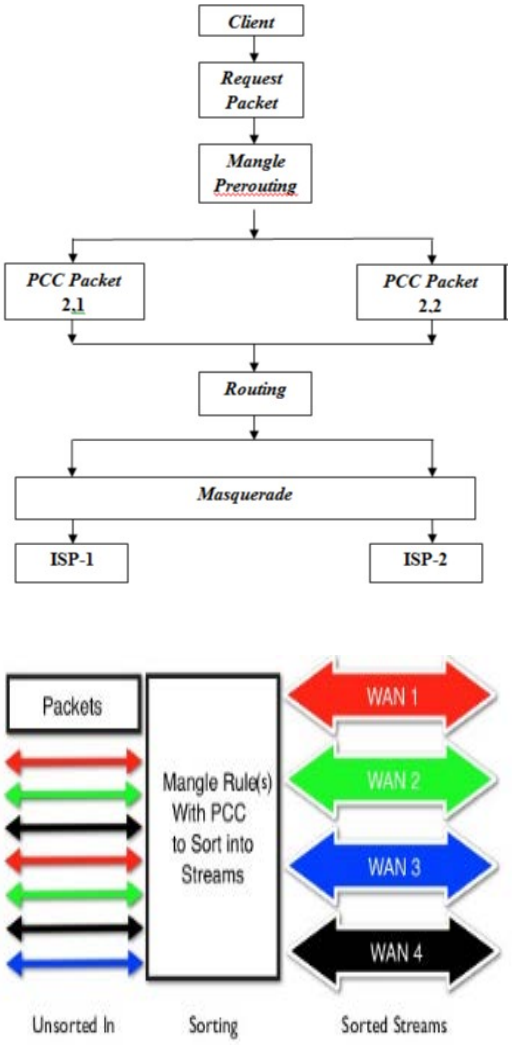


Fig. 2. PCC algorithm process.

The advantage of PCC is the relationship between client and server is intact because it always passes the same path. For example, when we access YouTube, if we use the Round Robin method, the IP is always changing, whereas if using PCC is not, it minimizes the lag when browsing or downloading [24].

The PCC method is a more optimal load balance method because it can perform the function of sharing traffic loads become two ISP [25]. The weakness of this

method is when the load balance process runs. It prioritizes cache/history on certain ISP paths. This condition means there will be a build-up of paths on an ISP because the paths used by other ISPs are congested.

Figure 3 shows this study uses a proxy router where the router functions to implement and test the PCC algorithm. There is also a server for monitoring to analyse traffic when implementing the PCC algorithm. Two ISP providers are where clients access the internet when carrying out load balances. And ten clients to test the PCC algorithm when the client accesses the internet. Figure 4 also shows a description of the PCC algorithm that occurs in the proxy router.

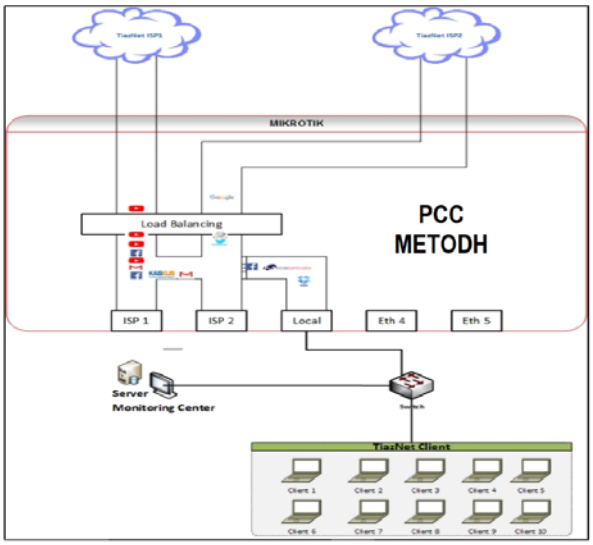


Fig. 3. Algorithm diagram network of PCC.

```
[admin@MikroTik] > interface
[admin@MikroTik] /interface> set name=ISP1 ether1
[admin@MikroTik] /interface> set name=ISP2 ether2
[admin@MikroTik] /interface> set name=Local ether3
[admin@MikroTik] /interface> /ip address
[admin@MikroTik] /ip address> add address=192.168.1.2/24 interface=ISP1
[admin@MikroTik] /ip address> add address=192.168.42.1/24 interface=ISP2
[admin@MikroTik] /ip address> add address=192.168.88.1/24 interface=Local
[admin@MikroTik] /ip address> /ip firewall mangle
[admin@MikroTik] /ip firewall mangle> add chain=input in-interface=ISP1 action=mar
[admin@MikroTik] /ip firewall mangle> add chain=input in-interface=ISP1 action=mark-connection new-connection-mark=Koneksi1
[admin@MikroTik] /ip firewall mangle> add chain=input in-interface=ISP2 action=mark-connection new-connection-mark=Koneksi2
[admin@MikroTik] /ip firewall mangle> add chain=output connection-mark=Koneksi1 action=mark-routing new-routing-mark=Gateway1
[admin@MikroTik] /ip firewall mangle> add chain=output connection-mark=Koneksi2 action=mark-routing new-routing-mark=Gateway2
[admin@MikroTik] > /ip firewall mangle
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[admin@MikroTik] /ip firewall mangle> add chain=prerouting dst-address=192.168.42.0/24 action=accept in-interface=Local
[admin@MikroTik] /ip firewall mangle> add chain=prerouting dst-address-type=!local in-interface=Local per-connection-classifier=
ection new-connection-mark=Koneksi1 passthrough=yes
[admin@MikroTik] /ip firewall mangle> add chain=prerouting dst-address-type=!local in-interface=Local per-connection-classifier=
ection new-connection-mark=Koneksi2 passthrough=yes
[admin@MikroTik] /ip firewall mangle> add chain=prerouting connection-mark=Koneksi1 in-interface=local action=mark-routing new-r

[admin@MikroTik] /ip firewall mangle> add chain=prerouting connection-mark=Koneksi2 in-interface=local action=mark-routing new-r
```

Fig. 4. Algorithm configuration of PCC.

2.2. Round Robin (RR)

Round Robin method, or commonly called Round Robin, is a method on load balance using the algorithm. Round Robin forms a particular series, which will be used as a queue system in the mangle rule that will be formed [4, 26]. The Round Robin algorithm focuses on justice. Robin Round uses the ring as a queue for storing packets. Each packet in the queue has the same execution time and will be executed alternately. If the packet cannot be completed in turn, the packet will be saved back to the queue, waiting for the next turn. The advantage of the RR algorithm is that each job will be carried out alternately, and they do not have to wait for the previous one to be completed. But if the packet load is large, then the Round Robin will take a long time to complete all the work. The Round Robin disadvantage is that the largest packet takes a long time to complete [27].

Figure 5 is an algorithm and process from the Round Robin method by distributing traffic loads evenly on two connection lines when traffic usage traffic occurs. Round Robin also divides the load evenly and turns by using the gateway simultaneously. Round Robin can be used as a backup connection if one of the connections down the algorithm has been designed to do auto-failover. Round Robin also divides the load evenly and turns by using the gateway simultaneously. The advantage of it is when the Round Robin method is busy. The Round Robin algorithm will distribute the load evenly on the two connection lines. Still, with a path that is not always the same, this is a weakness of the Round Robin method because when the connection between the server and the client is not still the same, then when browsing, downloading, or streaming can lag [28-30].

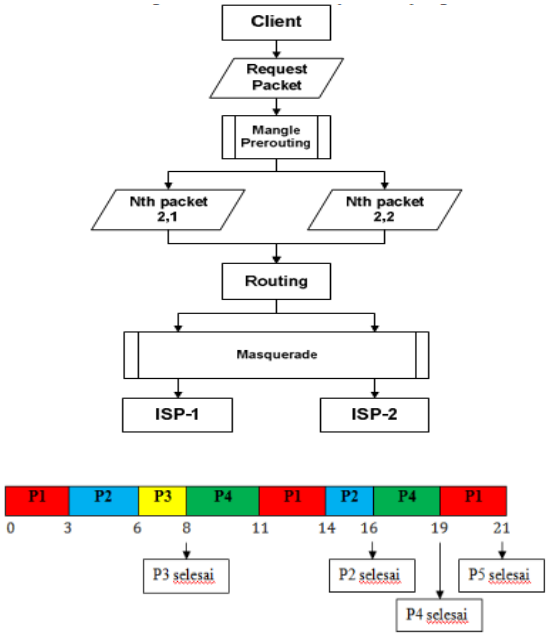


Fig. 5. The methods of round robin algorithm.

Round Robin method is a load balance method whose function is almost the same as the PCC method, which divides the load of both ISPs so that traffic is more

optimal. However, it does not care about the destination, whether it passes the same or different route from the previous route. Round Robin will still divide the queue equally between the two ISP lines; consequently, when we access the internet for destination needs similar to before, the path does not use cache history. As a result, the IP is always changing so that traffic is not optimal if we play the game or open YouTube; there is the term buffering or lag that will happen.

Figure 6 shows the Round Robin method also uses a proxy router where the router functions to implement Round Robin algorithm testing and implementation. Figure 7 shows the Round Robin method also uses a proxy router where the router functions to perform Round Robin algorithm testing. There is also a server for monitoring to analyse traffic when implementing the Round Robin algorithm. Two ISP Providers are where clients access the internet when carrying out load balances. There are also ten clients to test the Round Robin algorithm when the client accesses the internet.

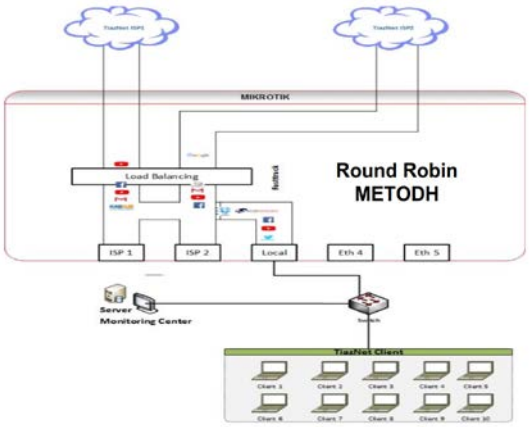


Fig. 6. Round Robin algorithm network diagram.

```
[admin@MikroTik] > /interface
[admin@MikroTik] /interface> set name=ISP1 ether1
[admin@MikroTik] /interface> set name=ISP2 ether2
[admin@MikroTik] /interface> set name=Local ether3
[admin@MikroTik] /interface> /ip address
[admin@MikroTik] /ip address> add address=192.168.1.2/24 interface=ISP1
[admin@MikroTik] /ip address> add address=192.168.42.179 interface=ISP2
[admin@MikroTik] /ip address> add address=192.168.88.1/24 interface=Local
[admin@MikroTik] /ip address> add address=192.168.42.179/24 interface=ISP2
[admin@MikroTik] /ip address> /ip firewall mangle
[admin@MikroTik] /ip firewall mangle> add chain=prerouting in-interface=Local conn
[admin@MikroTik] /ip firewall mangle> add chain=prerouting in-interface=Local connection-state=new nth=2,1
[admin@MikroTik] /ip firewall mangle> add chain=prerouting in-interface=Local connection-state=new nth=2,1 action=mark-connection new-connection-mark=Koneksi1 p
s
[admin@MikroTik] /ip firewall mangle> add chain=prerouting in-interface=Local connection-mark=Koneksi1 action=mark-routing new-routing-mark=Koneksi1 passthrough
[admin@MikroTik] /ip firewall mangle> add chain=prerouting in-interface=Local connection-state=new nth=2,2 action=mark-connection new-connection-mark=Koneksi2 p
s
[admin@MikroTik] /ip firewall mangle> add chain=prerouting in-interface=Local connection-mark=Koneksi2 action=mark-routing new-routing-mark=Koneksi2 passthrough
```

Fig. 7. Round Robin algorithm configuration.

3. Peer Classifier Robin (PCR) Design

Peer Classifier Robin (PCR) is a method that was born from a hybrid Peer Classifier Connection and Round Robin method. By integrating both algorithms and slightly modifying the algorithm so that it is expected to be able to minimize weaknesses that exist in the Peer Connection Classifier (PCC) and the Round Robin methods.

The main objective of PCR implementation is to minimize some weaknesses of the PCC and Round Robin methods and optimize several advantages of both. By using this method, it is expected that there will be no traffic congestion in one path and IP changes when accessing the internet. Thus, when accessing the internet, traffic will prioritize the cache of destination destinations, but if the route is full, then the traffic will move on the current path blank.

Figure 8 shows the Peer Classifier Robin (PCR) method diagram network. There is also a server for monitoring to analyse traffic when implementing the Peer Classifier Connection algorithm. Two ISP Providers are where clients access the internet when carrying out load balances. There are also ten clients to test the proposed algorithm when the client accesses the internet. Figure 9 shows the configuration network of the Peer Classifier Robin (PCR) method.

4. Results and Discussion

Several testings are performed as the implementation of several network configurations. Each network configuration has 2 ISP providers, one proxy device, one switch device, one server, and ten clients. Three network configurations, namely: a PCC, a Round Robbin, and a PCR, have been designed, as shown in Figs. 1-3, respectively. The discussion of each test is discussed below.

Figures 10 and 11 show visual traffic were accessing the internet traffic conditions are not balanced between ISP1 and ISP2, traffic is more dominant at ISP2.

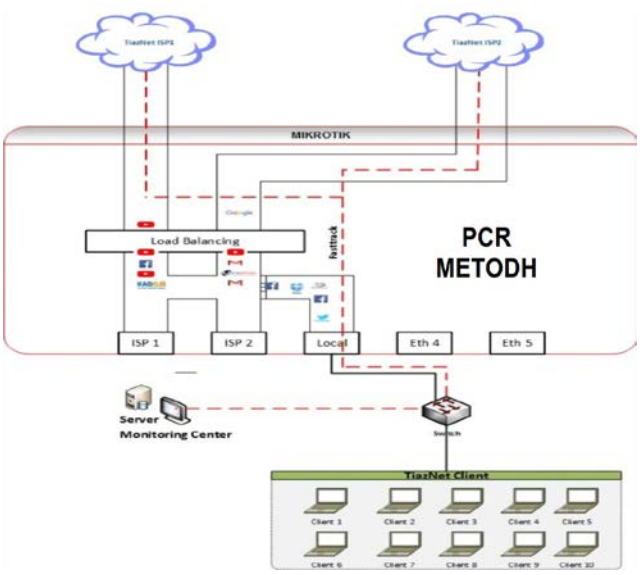
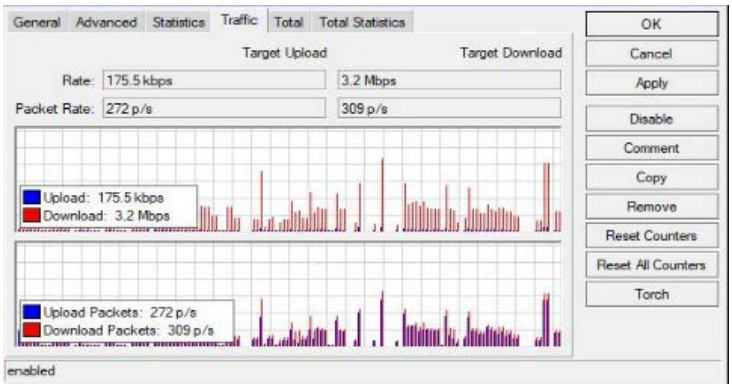


Fig. 8. Diagram network of the Peer Classifier Robin algorithm.

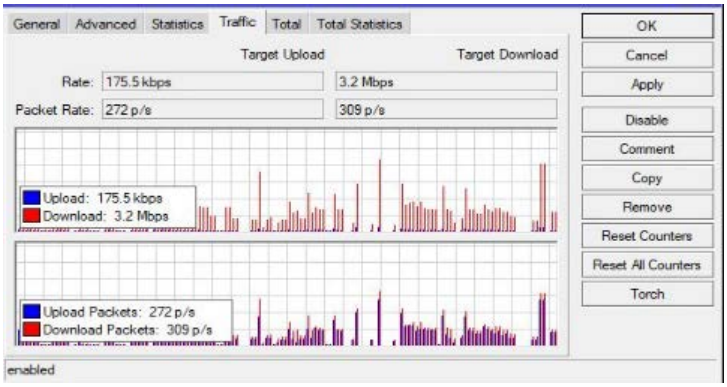
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[admin@MikroTik] /ip address> add address=192.168.88.1/24 interface=Local
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[admin@MikroTik] /ip firewall mangle> add chain=input in-interface=ISP1 action=mar
[admin@MikroTik] /ip firewall mangle> add chain=input in-interface=ISP1 action=mark-connection new-connection-mark=Koneksi1
[admin@MikroTik] /ip firewall mangle> add chain=input in-interface=ISP2 action=mark-connection new-connection-mark=Koneksi2
[admin@MikroTik] /ip firewall mangle> add chain=output connection-mark=Koneksi1 action=mark-routing new-routing-mark=Gateway1
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ection new-connection-mark=Koneksi2 passthrough=yes
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[admin@MikroTik] /ip firewall mangle> add chain=prerouting connection-mark=Koneksi2 in-interface=local action=mark-routing new-r
[admin@MikroTik] /ip address> /ip firewall mangle
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[admin@MikroTik] /ip firewall mangle> add chain=prerouting in-interface=Local connection-mark=Koneksi1 action=mark-routing new-routing-mark=Koneksi1 passthrough
[admin@MikroTik] /ip firewall mangle> add chain=prerouting in-interface=Local connection-state=new nth=2,2 action=mark-connection new-connection-mark=Koneksi2 p
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[admin@MikroTik] /ip firewall mangle> add chain=prerouting in-interface=Local connection-mark=Koneksi2 action=mark-routing new-routing-mark=Koneksi2 passthrough
```

Fig. 9. Configuration network of the peer classifier robin algorithm.



(a) ISP 1.



(b) ISP 2.

Fig. 10. The traffic of PCC.

Figure 11 shows the traffic when accessing YouTube as many as three pieces and accessing streaming net tv as much as one gateway path, which is determined by an algorithm that is based on cache history that is still related. It means that the 3 YouTube packets are passed through the ISP 2 line where we can see there is a build-up in the path, while the streaming net tv is passed through the ISP 1 line so that the traffic runs stable. However, the three YouTube pieces do not experience buffering because the relationship between the client and server is guaranteed because the packet is passed the same gateway path when the packet accessed is still interrelated so that the cache history is always stored in the gateway.

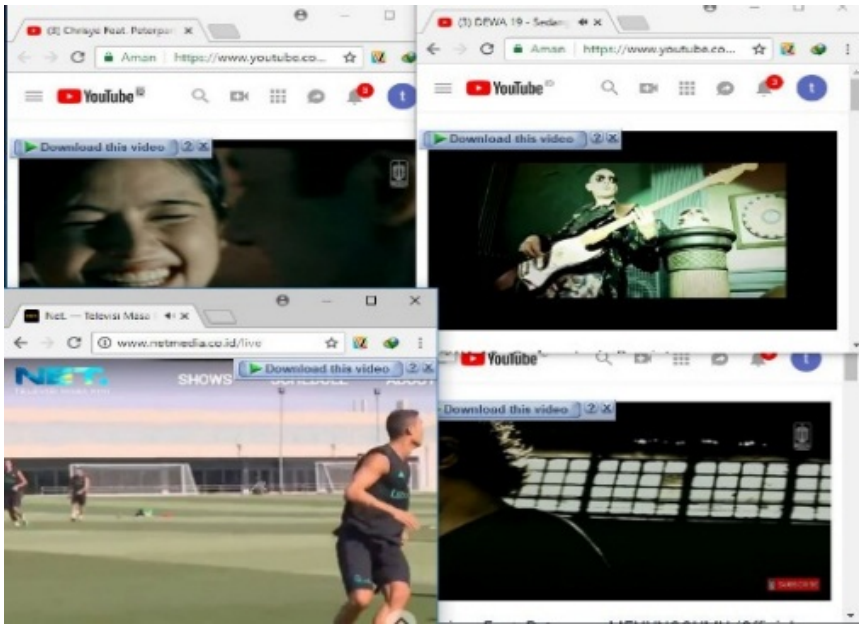


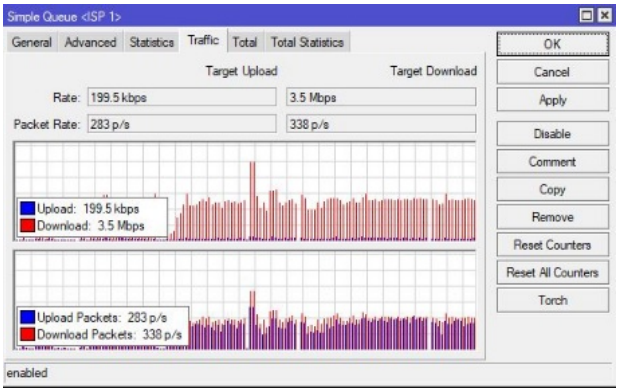
Fig. 11. The use of PCC packets load balance algorithm.

After that, the QoS calculation is done using the Wireshark application, which is carried out using the PCC method. The QoS parameters are calculated as packet loss, delay time, and throughput. Figure 12 shows the QoS parameters result.

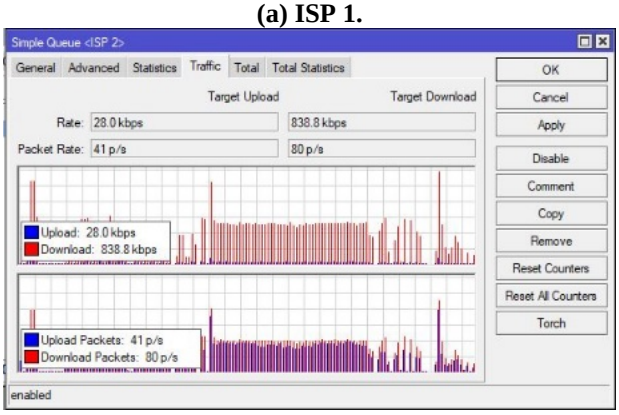
Time	Source	Destination	Protocol	Info	Time delta fr	Length	Frame len	Identifica	Identification
341.776882	114.124.224.13	172.168.1.253	GQUIC	Payload (Encryp	0.003727	1392	1392	0x0000 (0)	0x0000 (0)
341.778802	110.50.83.35	172.168.1.253	TCP	443 > 60504 [A	0.00192	1506	1506	0x2426 (9	0x2426 (9254)
341.78195	172.168.1.253	114.124.224.13	GQUIC	Payload (Encryp	0.003148	85	85	0x6dbd (2	0x6dbd (28093)
341.782742	74.125.164.70	172.168.1.253	GQUIC	Payload (Encryp	0.000792	1392	1392	0x0000 (0)	0x0000 (0)
341.782862	172.168.1.253	74.125.164.70	GQUIC	Payload (Encryp	0.00012	85	85	0x6dbe (2	0x6dbe (28094)
341.784776	110.50.83.35	172.168.1.253	TCP	[TCP Previous s	0.001914	1506	1506	0x2428 (9	0x2428 (9256)
341.785694	172.168.1.253	110.50.83.35	TCP	60504 > 443 [A	0.000918	66	66	0x6dbf (28	0x6dbf (28095)
341.787891	74.125.164.70	172.168.1.253	GQUIC	Payload (Encryp	0.002197	1392	1392	0x0000 (0)	0x0000 (0)
341.788806	172.168.1.253	74.125.164.70	GQUIC	Payload (Encryp	0.000915	85	85	0x6dc0 (28	0x6dc0 (28096)
341.7907	110.50.83.35	172.168.1.253	TCP	443 > 60504 [A	0.001894	1506	1506	0x2429 (9	0x2429 (9257)
341.791616	172.168.1.253	110.50.83.35	TCP	[TCP Dup ACK 1	0.000916	66	66	0x6dc1 (28	0x6dc1 (28097)
341.793392	74.125.164.70	172.168.1.253	GQUIC	Payload (Encryp	0.001776	1392	1392	0x0000 (0)	0x0000 (0)
341.794313	172.168.1.253	74.125.164.70	GQUIC	Payload (Encryp	0.000921	85	85	0x6dc2 (28	0x6dc2 (28098)
	Packet Loss	Packet Data Dikirim - Packet Data Diterima (Y)	0 %						
	Througput	Total Length =	828446152						
		Total Length / Time =	1656892.304						
		Througput =	1656.892304						
			1.656892304	mbps					
	Delay Time	Avg (Length/Link Bandwith) =	0.005645879						
			5.645879133	ms					

Fig. 12. The QoS parameters results of PCC.

Figure 13 shows that the Round Robin algorithm makes traffic run in balance on both gateway lines. But the problem is that each packet accessed is not always passed the same packet when accessing the internet even though the packet is interrelated. This results in the buffering of one of the accessed YouTube. The use of traffic is tested when accessing the internet to streaming sites, as can be seen in Fig. 14.



(a) ISP 1.



(b) ISP 2.

Fig. 13. Traffic Round Robin.

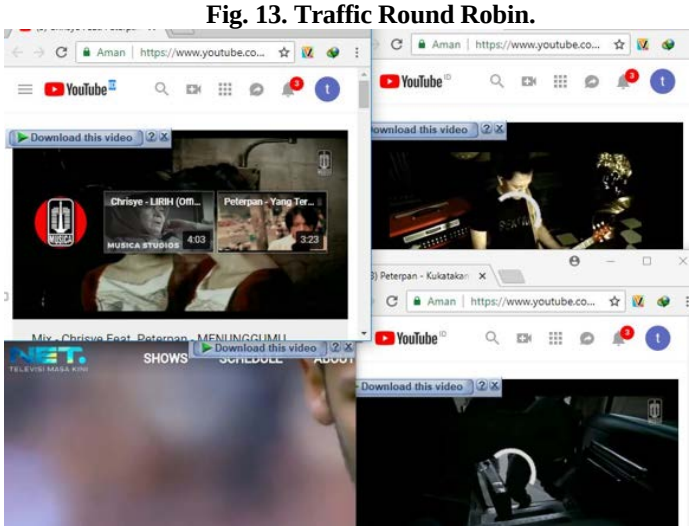


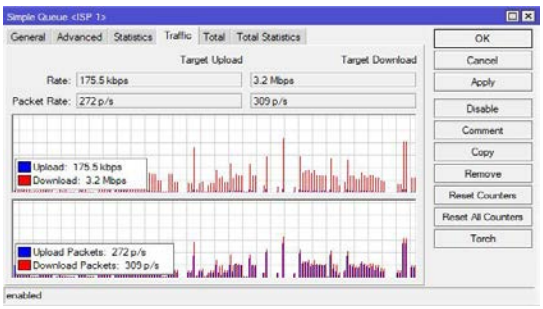
Fig. 14. The use of the Round Robin packet load balance algorithm.

The QoS parameters for the Round Robin network configuration are shown in Fig. 15.

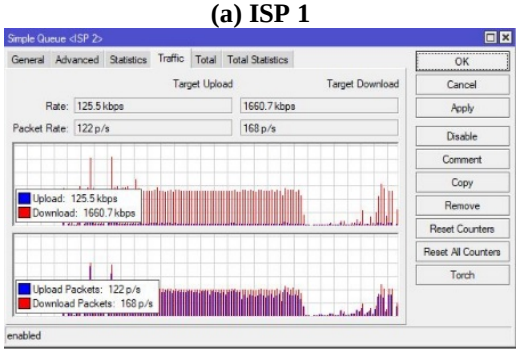
Time	Source	Destination	Protocol	Length	Time delta from previou	Frame length	Info
662.478213	114.120.225.12	192.168.43.2	GQUIC	1392	0.000241	0.002624	1392 Payload (Encrypted), PKN: 15
662.481078	114.121.194.83	192.168.43.2	GQUIC	1392	0.002865	0.003695	1392 Payload (Encrypted), PKN: 66
662.487638	192.168.43.2	114.120.225.12	GQUIC	81	0.00656	-0.0056	81 Payload (Encrypted), PKN: 199
662.488601	192.168.43.2	114.121.194.83	GQUIC	89	0.000963	0.000763	89 Payload (Encrypted), PKN: 75,
662.490327	114.121.194.83	192.168.43.2	GQUIC	76	0.001726	0.007718	76 Payload (Encrypted), PKN: 67
662.499771	114.120.225.12	192.168.43.2	GQUIC	1392	0.009444	-0.00651	1392 Payload (Encrypted), PKN: 16
662.502703	114.121.194.83	192.168.43.2	GQUIC	1369	0.002932	-0.00197	1369 Payload (Encrypted), PKN: 68
662.503668	114.120.225.12	192.168.43.2	GQUIC	1392	0.000965	0.001192	1392 Payload (Encrypted), PKN: 17
662.505825	114.121.194.83	192.168.43.2	GQUIC	1392	0.002157	0.000999	1392 Payload (Encrypted), PKN: 69
662.508981	114.121.194.83	192.168.43.2	GQUIC	1392	0.003156	-0.00313	1392 Payload (Encrypted), PKN: 70
662.509009	114.120.225.12	192.168.43.2	GQUIC	1392	0.000028	0.001004	1392 Payload (Encrypted), PKN: 18
662.510041	192.168.43.2	114.121.194.83	GQUIC	84	0.001032	-0.00103	84 Payload (Encrypted), PKN: 76,
		Packet Loss	Packet Data Dikirim - Packet Data Diterima (Y)	0 %			
		Througput	Total Length =	628446152			
			Total Length / Time =	949314.429			
			Througput =	949.314429			
				0.949314429 mbps			
		Delay Time	Avg (Length/Link Bandwith) =	0.002067 ms			
				2.067			

Fig. 15. The QoS parameters results of Round Robin.

Finally, Fig. 16 shows the PCR performances. Based on those figures, the PCR algorithm runs in a balance between ISP 1 and ISP 2, but the determination of the gateway path is prioritized according to the packet. It is still interconnected when accessing the internet. The situation means that it can minimize buffering, but the packet can also run in a balanced manner on two gateway lines. The problem in this algorithm is still often a connection breakdown; the traffic above can be seen happening at ISP 2. The use of traffic is tested when accessing the internet to streaming sites, as can be seen in Fig. 17.



(a) ISP 1



(b) ISP 2

Fig. 16. Traffic PCR

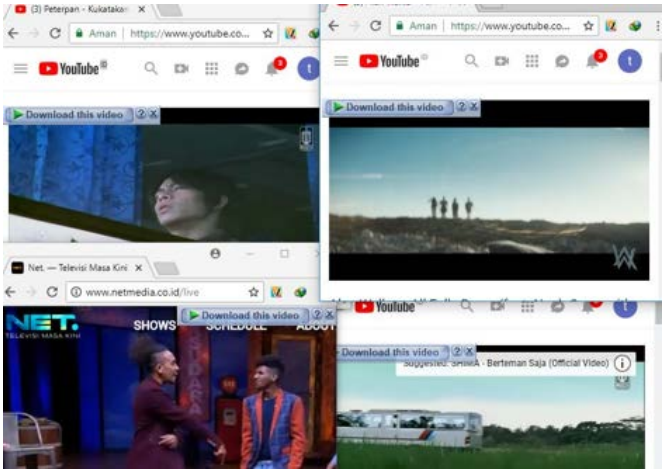


Fig. 17. Using the PCR packet load balance algorithm.

Also, the QoS parameters for the PCR network configuration are shown in Fig. 18.

Time	Source	Destination	Protocol	Info	Time delta	Length	Frame len	Identifica	Identification
321.31751	192.168.1.2	173.194.22.40	GQUIC	Payload (En	0.006895	102	102	0x4dc6 (19	0x4dc6 (19910)
321.318393	173.194.22.40	192.168.1.2	GQUIC	Payload (En	0.000883	1392	1392	0x0000 (0)	0x0000 (0)
321.319518	209.85.229.252	192.168.1.2	GQUIC	Payload (En	0.001125	1392	1392	0x0000 (0)	0x0000 (0)
321.320403	173.194.22.40	192.168.1.2	GQUIC	Payload (En	0.000885	1392	1392	0x0000 (0)	0x0000 (0)
321.321535	192.168.1.2	173.194.22.40	GQUIC	Payload (En	0.001132	102	102	0x4dc7 (19	0x4dc7 (19911)
321.327118	192.168.1.2	209.85.229.252	GQUIC	Payload (En	0.005583	83	83	0x4dc8 (19	0x4dc8 (19912)
321.328032	192.168.1.2	209.85.229.252	GQUIC	Payload (En	0.000914	78	78	0x4dc9 (19	0x4dc9 (19913)
321.32883	209.85.229.252	192.168.1.2	GQUIC	Payload (En	0.000798	1375	1375	0x0000 (0)	0x0000 (0)
321.331711	173.194.22.40	192.168.1.2	GQUIC	Payload (En	0.002881	1392	1392	0x0000 (0)	0x0000 (0)
321.332584	192.168.1.2	173.194.22.40	GQUIC	Payload (En	0.000873	104	104	0x4dca (19	0x4dca (19914)
321.333558	173.194.22.40	192.168.1.2	GQUIC	Payload (En	0.000974	1392	1392	0x0000 (0)	0x0000 (0)
321.335573	173.194.22.40	192.168.1.2	GQUIC	Payload (En	0.002015	1392	1392	0x0000 (0)	0x0000 (0)
321.336372	209.85.229.252	192.168.1.2	GQUIC	Payload (En	0.000799	1392	1392	0x0000 (0)	0x0000 (0)
321.337189	173.194.22.40	192.168.1.2	GQUIC	Payload (En	0.000817	1392	1392	0x0000 (0)	0x0000 (0)
321.337973	192.168.1.2	173.194.22.40	GQUIC	Payload (En	0.000784	118	118	0x4dcb (19	0x4dcb (19915)
321.34356	192.168.1.2	173.194.22.40	GQUIC	Payload (En	0.005587	106	106	0x4dcc (19	0x4dcc (19916)
321.344348	209.85.229.252	192.168.1.2	GQUIC	Payload (En	0.000788	1392	1392	0x0000 (0)	0x0000 (0)
321.346599	173.194.22.40	192.168.1.2	GQUIC	Payload (En	0.002251	1392	1392	0x0000 (0)	0x0000 (0)
321.348925	173.194.22.40	192.168.1.2	GQUIC	Payload (En	0				
	Packet Loss	Packet Data Dikirim - Packet Data Diterima (Y)	0.05	%					
	Througput	Total Length =	928446152						
		Total Length / Time =	2321115.38						
		Througput =	2321.11538						
			2.32111538	mbps					
	Delay Time	Avg (Length/Link Bandwith) =	0.001943						
			1.943	ms					

Fig. 18. The QoS parameters results of PCR.

Table 1 shows a calculation of QoS on packets that are running when a user accesses the internet. Measurements are made using a Wireshark application and calculations using the QoS pad formula generally.

Table 1. QoS parameter calculations.

Measurement Method	Packet Loss (%)	Delay Time (ms)	Throughput (Mbps)
PCC	0	5.64	1.65
Round Robin	0	2.06	0.94
PCR	0.05	1.94	2.32

The table shows the results of calculating the QoS parameter when the packet runs on each method. And each method has advantages and disadvantages of each based on QoS measurements during testing. In Fig. 19, it can also be seen as a comparison of the calculations of Wireshark three methods that have been tested in the form of diagrams.

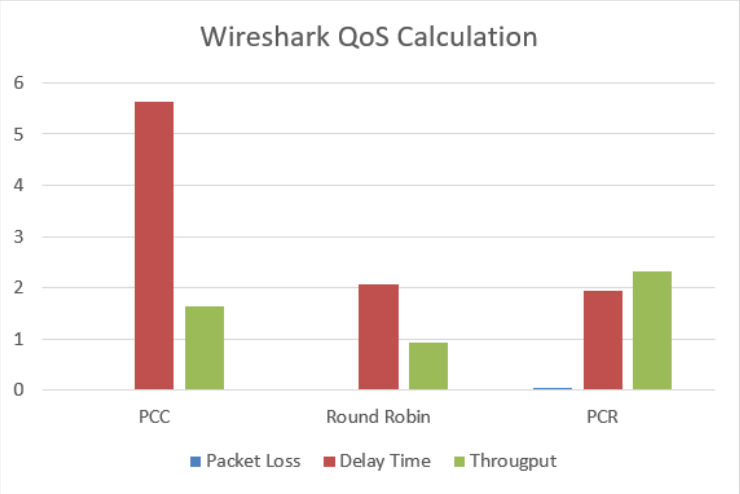


Fig. 19. Measurement chart of the load balance method.

5. Conclusion

Peer Classifier Robin (PCR) is a hybrid method from the Peer Connection Classifier and Round Robin methods. The PCR has the optimal method for implementing load balance, by reducing latency (1.94 ms) and increasing throughput (2.32 Mbps) when used in accessing the internet. Thus, the PCR method can minimize weaknesses in the Peer Connection Classifier and Round Robin methods.

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